Venkata Sai Sreeharsha Kotta

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Summary

Machine Learning Engineer with 5+ years of experience productionizing ML, deep learning, and LLM/VLM systems at both a **Fortune 500 enterprise** and an **early-stage startup**. At **HP**, built forecasting models and data pipelines on Databricks and PySpark for global supply-chain operations. At **Vinci**, joined as the first ML engineer and owned a real-time video AI platform end to end, from **VLM serving with vLLM** and edge GPU optimization to backend, CI/CD, and infrastructure. Effective in structured enterprise teams and fast-moving startups alike, taking systems from prototype to production.

Education

- **Georgia State University** — Atlanta, USA
  Master of Science in Analytics; Data Science and Analytics; GPA: 4.0/4.0 — *Aug. 2022 - May 2024*
- **Shiv Nadar University** — Delhi, India
  Bachelor of Technology in Computer Science and Engineering; Minor in Management — *Aug. 2016 - Jun. 2020*

Experience

- **Vinci Education Corporation** — Dunn Loring, VA, USA
  Senior Machine Learning Engineer — *May 2024 – Present*
  Early-stage startup; first ML engineer with end-to-end ownership across ML, backend, and infrastructure. Led one ML engineer, working alongside a PM and a web developer.

  AI Systems & VLM Serving
  - Led end-to-end design and **productionization** of a real-time detection platform scaled to **100+ edge devices and 600+ cameras** across multiple sites, using an **edge-first MicroK8s** architecture for on-device inference and **AWS EKS** for the web app, messaging/streaming, and operational workflows.
  - Designed a multi-model video-understanding pipeline, **evaluating and integrating pretrained and vendor models** (**YOLO**, pose estimation, **3D CNN** RGB/motion, transformer temporal models, and larger **vision-language models/LLMs**) to classify abuse vs non-abuse and generate scene descriptions.
  - Designed a **pose → VLM cascade** for fall detection: **YOLO-Pose** flags candidate events, then 6 evenly spaced frames are tiled into a grid so an image VLM (**Qwen-VL**) can reason over time, returning fall probability, label, and scene description.
  - Scaled cloud VLM inference for **600+ cameras** on **4 AWS GPUs** by serving **Qwen-VL** and other open-source VLMs with **vLLM** and batching requests across cameras; split 10s clips into 3s snippets with a **fast-check → detailed-analysis** cascade, bringing average record-to-alert time from **∼20s to 5–6s** in real time across all sites.
  - Achieved **95%+ precision and 92% recall in production** across multiple event labels (video and audio), tracked on **live dashboards across all deployed sites**, through **feature engineering, threshold tuning, temporal smoothing, and observability**, processing **∼9,000 segments per camera** while surfacing only **1-2 actionable alerts**.
  - Cut edge alert latency from **6–7s to <2s** at **25 FPS** on **NVIDIA GPUs** by converting **YOLO-Pose and Qwen-VL from FP32 to FP16**, batching pose inference, frame skipping, and moving VLM description generation off the alert path.
  - Built a **semantic video-search** capability with large vision-language models, enabling natural-language retrieval across all cameras and recorded footage for rapid incident review.
  - Instrumented LLM/VLM pipelines with **Langfuse tracing** (per-request latency, token usage, prompts and outputs) to debug slow or incorrect VLM calls across the inference queue.

  MLOps & Data
  - Orchestrated, scaled, and hardened **model training and serving pipelines** with **SageMaker Pipelines**, **AWS Lambda + S3**, **model registry/versioning**, and **CI gating tests**, promoting only validated models across all customers and devices.
  - Implemented monthly **auto-retraining loops** with **model monitoring and drift detection** from production feedback, plus a **privacy-preserving face-blurring pipeline** for secure, compliant data handling.
  - Curated and **versioned** a **7K+** labeled clip dataset with QA and augmentation, ensuring reproducible training and consistent feature/label governance.

  Platform & Infrastructure (owned end to end)
  - Developed and optimized the cloud platform on **AWS EKS** (**Spring Boot APIs, Angular, SQL**), reaching **sub-millisecond server-side API and database query latency**, with **Nginx + cert-manager** ingress and **S3 data-lake pipelines** for media storage and reporting.
  - Built **CI/CD and GitOps from scratch**: self-hosted **GitLab runners** and pipelines with automated tests, **Trivy** image scanning, and **Renovate** dependency updates; **ArgoCD** managing edge devices and **dev/test/prod EKS clusters**; **External Secrets** with **AWS Secrets Manager** and least-privilege **IAM**.
  - Ran edge-to-cloud messaging with **RabbitMQ and VerneMQ** for device communication, deployments, and status updates, and managed the edge fleet with **AWS Systems Manager**.

- Set up **observability** with **Prometheus, Grafana, and Loki** (camera activity, pod failures, pipeline health) plus reporting dashboards, and automated site onboarding to cut deployment time from **6–7 hours to <30 minutes** per site; **mentored junior engineers**.

- **Institute of Insight, Georgia State University** Atlanta, USA
  Graduate Research Assistant *Jan 2023 – May 2024*
  - Extracted and analyzed unstructured financial text at scale using Pushshift (multi-thread), PRAW, and web scraping across Reddit, news, and earnings-call transcripts; applied **NLP** to identify company names/tickers and surface supply-chain topics
  - Implemented textual analysis with seeded word embeddings (**Word2Vec**, **CBOW**) in Python to build an organization-level metric assessing supply-chain ethicality from web-scraped data.
  - Applied **Large Language Models** for content classification, labeling millions of tweets to detect mental-trauma signals at scale.
- **Hewlett Packard Inc** Bengaluru, India
  Data Scientist *Aug 2020 – Aug 2022*
  - Automated end-to-end **ETL, model training, and evaluation** via scalable data pipelines in **Azure Databricks** and **Azure Data Factory** using **Spark (PySpark, SparkR)** on distributed clusters, saving **120 labor hours monthly**.
  - Developed **time series forecasting** models with **machine learning and deep learning** in Python and R for Part Demand, Repair, Call Volumes, and Warranty Interventions, improving budget accuracy by **15%** and reducing forecasting errors by **20%**.
  - Forecasted spare-parts demand with **89% accuracy**, driving **$26M savings** by minimizing overstock and part wastage; led global field-service coverage optimization, boosting accuracy by **17%** via a Staff Planning Tool.
  - Built a **Spare Part Replacement Recommendation Engine** using **NLP and Transformer-based models** to help customer-service agents assist with damage and repair, reducing wait time to **15 seconds**.
  - Applied **feature engineering** and predictive analytics for operational risk and resource-allocation analysis, optimizing supply-chain resources and inventory management.
  - Published part-demand forecasting research at HP's internal **DSKD Summit** and delivered insights to cross-functional stakeholders through interactive **Power BI** dashboards.
- **Hewlett Packard Inc** Chennai, India
  Data Science Intern *Feb 2020 – Aug 2020*
  - Performed exploratory data analysis on monthly intervention snapshots to support long-term modeling with an expected ROI of **$6M** over 3 years.
  - Supported statistical and ML-based demand forecasting for inventory optimization and contributed to early pipeline automation using Python and R.

Projects

- **AI-Driven Video Summarization and Deepfake Synthesis**: Combined SSD face detection and Wav2Lip lip-sync with NLP tooling (Gensim, NLTK, spaCy, TF-IDF) for summarization, Tortoise TTS for voice conversion, and Google Speech Recognition; delivered an interactive Streamlit interface with attention to ethical use of deepfake technology.
- **Marketing Budget Optimization using Markov Chains**: Used R and the Channel Attribution package to model attribution paths and optimal budget allocation, comparing Markov Chains with heuristic models to improve channel evaluation and campaign performance.

Technical Skills

- **Languages**: Python, SQL, Scala, R, Java, Bash, PySpark
- **Data Engineering & Distributed**: Azure Databricks, Azure Data Factory, Spark (PySpark), Data Lakes (S3), ETL & Feature Pipelines, Streaming, REST APIs
- **Machine Learning & DL**: Computer Vision, Vision-Language Models (VLMs), LLMs, Transformers, CNNs, NLP, Time Series Forecasting, Feature Engineering, Model Optimization & Tuning, GenAI/LLM Fine-tuning
- **Libraries & Frameworks**: PyTorch, TensorFlow, Scikit-Learn, OpenCV, HuggingFace Transformers, vLLM, LangChain, Pandas, NumPy
- **MLOps & Deployment**: Docker, Kubernetes (MicroK8s, EKS), Helm, ArgoCD (GitOps), SageMaker Pipelines, Model Registry/Versioning, GitLab CI/CD (self-hosted runners), Git, Trivy, Renovate, AWS ECR/Lambda, External Secrets
- **Cloud Platforms**: AWS (EC2, EKS, S3, Lambda, SageMaker, ECR, CloudWatch, Systems Manager, Secrets Manager, IAM), Azure (Databricks, ADF)
- **Databases & Streaming**: PostgreSQL, MySQL, MongoDB, DynamoDB, RabbitMQ, VerneMQ
- **Monitoring & Observability**: Langfuse, Prometheus, Grafana, Loki, Power BI, Tableau

Publication

- Object and Currency Detection with Audio Feedback for Visually Impaired. In IEEE Region 10 Symposium (TENSYMP), pp. 1152-1155. IEEE, 2020